Customer Churn Analysis for Telecom Industry

Introduction: Customer churn refers to customers leaving a telecom service provider. Reducing churn is important because retaining customers is more cost-effective than acquiring new ones.

Abstract: This project analyzes telecom customer data to predict churn and identify the major factors influencing customer retention. Machine learning techniques were used to classify customers and generate business insights.

Tools Used: SQL, Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SHAP, Jupyter Notebook.

Steps Involved:
1. Data collection and loading.
2. Data cleaning and preprocessing.
3. Exploratory Data Analysis (EDA).
4. Feature engineering and encoding.
5. Logistic Regression model building.
6. Model evaluation using Accuracy, Precision, Recall, and F1-score.
7. SHAP analysis for explainability.
8. Customer segmentation into Loyal, At Risk, and Dormant groups.
9. Business recommendations for customer retention.

Conclusion: The project identified contract type, tenure, and monthly charges as key churn drivers. Predictive modeling and customer segmentation can help telecom companies improve retention, reduce churn, and increase long-term profitability.